Leakage-Safe Forecasting of Occupational Heat-Strain Risk from Wearable-Style Sensors: A Benchmark on Unseen Workers and Unseen Conditions

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***Abstract*—Occupational heat strain is a growing safety problem as extreme heat becomes more frequent and prolonged. Environment-only indices such as WBGT ignore individual response, while physiological indices often need core-temperature readings that are impractical on a worksite. Published machine-learning forecasters frequently split data by sample rather than by person, compare against few baselines, and depend on proxy labels rather than measured core temperature. We present a leakage-safe benchmark for forecasting heat-strain risk 15 minutes ahead on the open PROSPIE dataset (40 participants, 9 heat-stress scenarios, 17,264 minute-level records with rectal core temperature). Risk classes (Low below 38.0 °C, Moderate 38.0–38.5 °C, High 38.5 °C and above) come from measured core temperature, and core temperature, its baseline delta, and total sweat loss are removed from the inputs. Four wearable-plausible modalities (cardio-respiratory, skin-thermal, environmental, individual) are evaluated with Logistic Regression, Random Forest, XGBoost, and an attention-gated fusion network with an optional gradient-reversal branch. Under 8-fold participant-grouped validation, XGBoost reaches macro-F1 0.719 and the fusion network 0.674, against 0.219 for an always-Low predictor (accuracy 0.489). Restricting the fusion network to heart and breathing rate lowers macro-F1 to 0.426, although feature count and missing data confound this comparison and it is read cautiously. Leave-one-participant-out accuracy for XGBoost averages 0.736 (range 0.599–0.876); leave-one-condition-out accuracy averages 0.682, with indoor hot-factory work in full protective clothing the hardest case (0.43–0.51). Temperature scaling lowered calibration error for trees and logistic regression but raised it for the fusion network. We report these outcomes, including the lack of benefit from the adversarial branch, and set the result against four lines of recent heat-strain research — feature-importance forecasting of personal heat strain, adaptive physiological-strain indexing, a structured review of heat-stress indices, and climate-projected labour-capacity loss — to make explicit where a learned, multimodal forecaster adds information beyond fixed-form indices and where it does not.**

***Index Terms*—occupational heat strain, heat-stress forecasting, multi-sensor fusion, wearable sensors, subject-independent evaluation, core body temperature, calibration, PROSPIE**

I. Introduction

Heatwaves keep arriving earlier, lasting longer, and reaching higher peaks [22], [23], and outdoor manual labour absorbs a disproportionate share of the resulting harm. Construction, agriculture, mining, and front-line health work all put people in conditions where the body's own cooling system is the only thing standing between a normal shift and a medical emergency, and national injury statistics keep showing these trades near the top of heat-related fatality counts [22], [23]. Climate projections make the outlook worse rather than better: modelling the hours during which standard occupational heat-stress limits still permit safe work shows that window shrinking across several world regions over the coming decades [6]. Any system meant to protect a worker from heat illness has to say something useful before the person actually reaches a dangerous physiological state, not after.

Two broad families of tool exist today, and neither one is quite enough on its own. Environmental indices are cheap to deploy and well standardised — the Wet-Bulb Globe Temperature (WBGT, ISO 7243) is the most familiar example — but by construction they describe the air and radiant environment, not the person standing in it [4]. Two workers doing the same job in the same WBGT reading can carry very different strain depending on fitness, acclimatisation state, hydration, and how much protective clothing they have on, and a meta-analysis across more than two thousand outdoor workers found only modest correlations (around 0.4) between WBGT and the physiological measures it is meant to stand in for [10]. Physiologically grounded alternatives — the Predicted Heat Strain model of ISO 7933, and the Physiological Strain Index (PSI) family — respond to the person rather than just the environment, but at the cost of needing either invasive core-temperature measurement or assumptions about metabolic rate that are hard to satisfy continuously on a real job site [5], [4].

Machine learning has been applied to this gap with encouraging but incompletely reported results. Models trained on

commodity environmental and biometric streams have forecast tympanic or core temperature with mean absolute errors in the 0.03–0.11 °C range [1], [2], [3], which sounds precise until the evaluation protocol is examined closely. Several recurring weaknesses make these numbers hard to use as a basis for comparison. Train/test partitions are frequently drawn at the level of individual time-samples rather than people, so the same participant contributes rows to both sides of a split and the reported accuracy is optimistic [34], [33]. Baseline choices differ across studies, with few papers reporting both a strong learned model and a trivial reference predictor side by side, so a reader cannot always tell how much of a quoted number reflects genuine model skill rather than class imbalance. Ablation analyses that isolate which sensing channel actually carries predictive signal are uncommon, and calibration of the predicted class probabilities is rarely addressed at all. Evaluation, when it happens, typically stays inside a single laboratory protocol, leaving open whether a model fitted under one thermal scenario — a hot factory, say — still works in a different one, such as an outdoor desert shift or a humid indoor space.

Our position in this paper is that for this particular problem, careful evaluation is worth as much attention as architectural novelty. We build on the open PROSPIE dataset [20], which records rectal core temperature alongside several synchronised sensor streams across nine heat-stress scenarios, to construct a benchmark in which the label is defined from measured physiology, nothing the model sees at prediction time is derivable from that label, and every number we report comes from data on people, or scenarios, that the model has not encountered during training. The contributions of this work are as follows.

1) **Leakage-safe label and inputs.** Risk classes are derived directly from measured rectal temperature using action-limit thresholds. Core temperature, its trial-baseline delta, and the whole-trial sweat-loss scalar are excluded from every model input.
2) **A benchmark on unseen workers and unseen conditions.** Logistic Regression, Random Forest, XGBoost, and an attention-gated fusion network with a gradient-reversal option are compared under 8-fold participant-grouped validation, leave-one-participant-out (LOPO) testing over all 40 participants, and leave-one-condition-out (LOCO) testing over all 9 scenarios.
3) **Reference levels that match the class balance.** Because the Low class makes up 49% of windows, we report an always-Low predictor (accuracy 0.489, macro-F1 0.219) alongside the uniform-guess level, and we apply Holm correction to the paired tests.
4) **Sensor-modality evidence and its limits.** An ablation shows a large drop when the fusion network sees only heart and breathing rate, and we state which confounds prevent a firm causal reading.
5) **Full reporting of calibration and negative results.** Calibration error before and after temperature scaling is given for every model, including the case where scaling made the fusion network worse, and the adversarial branch is reported as providing no measurable benefit.

The rest of the paper is organised as follows. Section II reviews related work. Sections III and IV describe the dataset, the label, and the problem setup. Section V presents the models, and Section VI the evaluation protocol. Sections VII–XI report results, and Sections XII–XIV discuss implications, limitations, and conclusions.

II. Related Work

A. Heat-Stress and Heat-Strain Indices

Core body temperature sits at the centre of heat-strain assessment because it integrates the balance between metabolic heat production and the avenues of heat loss available to the body. When the surrounding air is warm, radiative and convective loss shrink and the body leans more heavily on evaporative cooling through sweat, a mechanism that is itself constrained whenever humidity is high or clothing blocks vapour transfer. The cardiovascular system compensates by routing more blood to the skin, which is why heart rate climbs and skin temperature rises as a shift goes on, and respiration quickens with workload. Heart rate, breathing rate, and skin temperature are therefore plausible wearable surrogates for core temperature, but they are imperfect surrogates precisely because each one is also shaped by fitness, hydration, posture, and clothing on top of thermal load [10], [12], [22].

A useful distinction, drawn out clearly in a recent structured review of the field, separates heat stress — the external thermal load placed on a person by the environment and the work — from heat strain, the body's internal physiological response to that load [4]. The review traces how that distinction has been operationalised historically. WBGT remains the most widely deployed stress index because a globe thermometer, a natural wet-bulb thermometer, and a dry-bulb thermometer are all that is needed to compute it, but field studies keep finding that it tracks physiological strain only loosely since it leaves out individual and clothing factors by design [4], [10]. Several lighter empirical indices sit alongside WBGT in practice — the Humidex and the Heat Index, both regression fits of perceived heat onto air temperature and humidity, and the Environmental Stress Index, which folds in solar radiation as a third term — and the review is useful precisely because it lines these formulas up against one another and shows how similar their inputs are even though their coefficients were fitted independently [4]. None of them, however, takes a measurement from the worker; they describe the air, not the body standing in it. On the other side of that divide sit the theoretical, physiologically grounded indices: the Predicted Heat Strain model of ISO 7933 integrates a bioheat balance over time using metabolic rate and clothing insulation to project the trajectory of core temperature and the maximum allowable exposure duration [4], [6], and the same review situates this family against the complementary standards that support it operationally — ISO 9886 for measuring strain directly on a person, ISO 7726 for

the instruments used to characterise the thermal environment, ISO 8996 for estimating metabolic rate, and ISO 9920 for estimating the thermal and evaporative resistance a given clothing ensemble adds [4]. Read together, these standards describe a measurement pipeline rather than a single number, and the review's contribution is less a new index than a clear map of which standard answers which question and where each one's assumptions start to break down, for example under extreme humidity or unusual postures.

Physiological composite indices occupy a middle position between the purely environmental and the purely theoretical. The Physiological Strain Index (PSI) proposed by Moran and colleagues combines a normalised heart-rate term with a normalised core-temperature term into a single zero-to-ten strain scale [5]. Its appeal is that it is simple and has been used across many studies, but a fixed critical core-temperature value baked into the formula is also its weak point: Buller et al. show that the standard PSI equation assumes a single maximal core temperature for everyone, and that this assumption produces strain scores that read as implausibly high for endurance athletes and implausibly low for soldiers working in fully encapsulating protective ensembles, even when both groups are, by direct physiological evidence, under comparably severe heat strain [5]. Their fix is to let the critical core-temperature term adapt to the gradient between core and skin temperature rather than holding it fixed, reasoning that a small core-to-skin gradient signals that the body's peripheral cooling pathway is already saturated and that cardiovascular strain is consequently higher than the core-temperature reading alone would suggest. In their validation, comparing eleven male and eight female endurance athletes against eight soldiers in protective equipment, the original PSI gave clearly different end-of-exercise scores between the two groups (about 11.4 versus 7.6, a gap the authors describe as statistically significant), while their adaptive version (aPSI) produced statistically indistinguishable scores for both groups (about 9.9 versus 9.0), consistent with both cohorts having reached a comparably severe strain state [5]. The broader lesson we take from that result is that a single fixed threshold, chosen once and applied uniformly, rarely transfers cleanly across the very different exercise-and-clothing combinations a real occupational dataset contains; this is part of our motivation for letting the fusion network in this paper learn per-modality weights from the data rather than fixing them in advance, and it is also why we treat threshold choice (Section IV-B) as an assumption to be stated plainly rather than a settled fact.

B. Multimodal Sensing and Fusion

Work on wearable physiological monitoring more broadly informs the architecture used here. One study combined gradient boosting with a Transformer encoder over electrodermal activity, blood-volume pulse, heart rate, skin temperature, and accelerometer channels, reaching 74.69% accuracy under a leave-one-subject-out protocol that is comparable in spirit to our LOPO setting; the same study flagged that raw accelerometer features can encode a wearer's identity well enough to hurt cross-person transfer, which is part of why the present work restricts motion-adjacent signal to rolling summary statistics rather than raw high-frequency channels [14]. A survey of AI-enhanced wearable sensing compares early, intermediate, late, and attention-based fusion strategies and observes that static concatenation of modalities tends to degrade whenever the reliability of an individual channel varies from sample to sample [15]; the attention gate used in our fusion network belongs to this last, adaptive category. A separate review centred on ECG-based stress recognition highlights a persistent gap between how models perform under controlled laboratory protocols versus free-living conditions [16], which is exactly the kind of shift our leave-one-condition-out protocol is designed to probe in a limited way. A hierarchical convolutional fusion of ECG and electrodermal signal reported roughly a four-point accuracy gain over non-hierarchical fusion for subject-independent stress detection [17], and a physics-informed, uncertainty-aware digital twin forecast core temperature in livestock with $R^2 = 0.783$ [18]. Neither of these last two studies targets human occupational heat strain directly, but both illustrate that combining a learned component with domain structure — hierarchy in one case, a physics prior in the other — tends to help more than naive concatenation, a pattern consistent with what the ablation in Section VIII of this paper finds for the attention gate relative to fixed equal weighting.

C. Machine-Learning Forecasting of Heat Strain

The line of work closest to ours is a sequence of studies by Seo, Choi, Koo and co-authors that apply interpretable machine learning directly to personal heat-strain forecasting in extremely hot environments. Their earlier study used permutation importance and SHAP values across Random Forest, XGBoost, SVR, and MLP regressors and found that four inputs — dry-bulb temperature, radiant temperature, relative humidity, and percentage of body fat — captured almost all of the achievable accuracy, with the full feature set lowering error by less than $0.047\,^{\circ}\mathrm{C}$ relative to that reduced set, a result the authors frame as a practical argument for lighter field data collection rather than exhaustive sensor deployment [1]. A companion study pushed further by adding cumulative working time as an explicit situational variable, reasoning that heat strain accumulates with continued exposure in a way that a snapshot of instantaneous conditions cannot capture. Their best model, an MLP trained with cumulative working time included, reached a mean absolute error of $0.034\,^{\circ}\mathrm{C}$ against measured tympanic temperature with $R^2 = 0.993$, a sizeable improvement over the MLP variant trained without that variable, and a permutation-importance analysis ranked cumulative working time as by some margin the most informative single feature, ahead of the environmental and anthropometric variables used in the earlier study [2]. A third paper from the same group coupled a heart-rate-driven variant of the Predicted Heat Strain model with a learned correction term and reported that the corrected model lowered the error in predicted maximum allowable exposure time from roughly five minutes to roughly twelve seconds relative to the uncorrected PHS prediction

[3]. Two things distinguish our setting from this body of work. First, their target is a continuous physiological quantity — a temperature or an exposure-time estimate — predicted and evaluated within a population that was also used for training and tuning, whereas we predict a three-level risk class and hold out people and conditions the model never saw during fitting, which is a harder and, we think, more decision-relevant test of generalisation. Second, where their strongest predictors are drawn from static anthropometry and cumulative exposure time, our benchmark keeps elapsed exposure time as a legitimate input (Section IV-A) precisely because the Seo et al. results make a strong case that it carries real signal, while still withholding every variable that is computed from, or that defines, the core-temperature label itself.

A separate strand of classification-oriented work is closer to our label structure. Zhao et al. built a composite strain index from core temperature, heart rate, and dehydration rate using principal component analysis, grouped the resulting index into five risk levels with K-means, and predicted the level from ambient conditions, work intensity, and duration using linear regression [11]. Their setting differs from ours in three respects: a single cohort of 56 young men with no cross-person validation split, modality weights fixed by the PCA decomposition rather than learned end-to-end, and a target defined at the current instant rather than forecast ahead in time. Because their target is a regression-derived composite index and ours is a forecast class defined directly from measured core temperature, we do not place the two sets of reported metrics side by side as if they were comparable.

The Physiological Strain Index, reviewed above alongside its adaptive aPSI extension [5], offers a second comparison point worth drawing out explicitly. Where aPSI adapts a single scalar threshold using the core-to-skin temperature gradient measured at that instant, our fusion network instead learns a soft attention weight over four whole sensing modalities, conditioned on the full feature vector rather than on one physiological gradient. The two approaches sit at different points on the same underlying idea — that a fixed, population-average threshold or weighting rarely survives contact with the range of clothing, fitness, and exposure conditions a real workforce presents — and the aPSI result that a single adapted parameter can equalise strain readings across two very differently equipped cohorts is, we think, encouraging evidence that the attention mechanism we use here is targeting a real source of variability rather than an artefact of our particular dataset. A direct, label-matched comparison between a fusion-network risk class and an aPSI-derived risk tier on the same PROSPIE trials would make this connection precise, and we list it as outstanding work in Section XIII.

D. Field and Population Studies

Several studies move the evaluation outside the laboratory and onto a working population or a projected future climate. Applying the Predicted Heat Strain model under future climate projections for six Australian sites spanning tropical, arid, and temperate zones, Hunt et al. estimated how long outdoor work could safely continue before a worker's modelled core temperature reached the ISO-recommended 38.0 °C limit, or alternatively before sweat loss reached the 5% body-mass dehydration limit, whichever constraint bound first [6]. Their projections, run under two representative concentration pathways (RCP4.5 and RCP8.5) and two future periods (2041–2060 and 2061–2080), show safe working duration falling by more than half at Port Macquarie and Griffith relative to current baseline conditions, with reductions of 20–50% further north, and they find that which of the two physiological limits binds first — core temperature or sweat loss — itself varies by location, with the sweat-loss constraint dominating at Clare and Griffith while the core-temperature constraint dominates elsewhere [6]. We see two implications for the present benchmark. First, the fact that the binding physiological constraint changes across sites is a reminder that core temperature, the single ground-truth signal PROSPIE provides, is a necessary but not sufficient description of heat strain; a fusion model trained only against a core-temperature-derived label cannot, by construction, learn anything about a sweat-loss-driven failure mode unless sweat rate is itself a legitimate real-time input, which in PROSPIE it is not (Section III-B). Second, Hunt et al.'s scenario-by-scenario variation in safe exposure time is conceptually close to our own leave-one-condition-out protocol, which likewise asks whether a model's behaviour under one combination of environment and workload generalises to a combination it has not seen; our finding that accuracy drops sharply for the fully protective-clothing scenario (Section IX-B) echoes their finding that the binding constraint, and therefore the safe outcome, is highly scenario-dependent rather than a single number that holds everywhere.

A randomised field study with Washington State farmworkers combined a tailored heat-education programme with a supervisor-facing decision-support application and measured strain using a heart-rate-derived core-temperature estimate fed through the PSI formula; perceived effort emerged as the strongest predictor of measured strain, and the intervention itself showed no clear association with self-reported symptoms, a result the authors read as an argument for objective physiological sensing over symptom self-report in this population [9]. A systematic review of heat strain among women workers catalogues environmental, organisational, and individual risk factors and concludes that individual factors, although they carry less aggregate weight than environmental ones in most published indices, are consistently under-represented relative to their actual contribution to risk [7]; this is part of why we keep an explicit individual-characteristics modality in the fusion network rather than folding age, body composition, and acclimatisation state into a single catch-all adjustment, even though Section VIII shows that modality's separate contribution has not yet been isolated from the others. A related occupational study of healthcare workers wearing personal protective equipment used in-ear and skin sensors inside a climate chamber and, through linear mixed-effects modelling, found additive effects of ambient heat and

protective-equipment use on ear temperature, heart rate, and mean skin temperature [8], a finding that lines up with our own observation that the fully protective-clothing scenario is the hardest case under leave-one-condition-out testing (Section IX-B).

E. Evaluation Methodology

Two methodological strands shape the protocol used here. First, when observations are clustered by person, a random split that ignores that clustering places correlated samples from the same individual on both sides of the train/test boundary and produces an optimistic estimate of how well the model will do on someone new; grouped splitting, in which every sample from a given person stays on one side of the split, is the standard remedy [34], and the broader point that information from a held-out set must never influence preprocessing choices made on the training set is a long-standing and still frequently violated rule in applied machine learning [37]. Second, with a modest number of independent groups — 40 participants or 9 conditions here — the estimate of generalisation error itself carries a wide interval, and the fold-wise scores that make up a cross-validated estimate are not independent of one another, which is precisely why naive paired tests across folds tend to be too permissive [33]. We follow these recommendations where the data allow it and flag the places where they do not: our confidence intervals (Table VII) are approximate rather than bootstrap-corrected, and the Holm-adjusted paired tests in Table V should be read as screening evidence rather than a final word on which model is best, exactly for the reasons Bates et al. and Tougui et al. set out.

F. Positioning

Table I summarises how the closest studies differ from the present benchmark in the properties discussed above. The entries are limited to what those studies state about their own design, and they describe differences in protocol rather than a ranking of quality; the studies pursue different goals, and our own work has the open items listed in Section XIII.

TABLE I
DESIGN PROPERTIES OF RELATED STUDIES

Study	Target	Evaluation focus
Seo et al. [1], [2]	Core / tympanic temp. (regression)	Feature importance; same-population evaluation
Choi et al. [3]	Max. allowable exposure time	PHS plus learned correction
Zhao et al. [11]	Five-level strain index	Single cohort, current instant
Alshamrani [14]	Four-class stress state	Leave-one-subject-out
Buller et al. [5]	Adaptive strain index	Cross-context threshold stability
This work	**3-class T_{re} risk at +15 min**	**LOPO, LOCO, ablation, calibration**

III. DATASET AND LABEL

A. The PROSPIE Dataset

We use the open-access PROSPIE dataset [20], which holds minute-resolution physiological and environmental recordings of 40 participants across 9 controlled heat-stress scenarios. The scenarios include a hot factory with protective equipment, a hot-humid indoor space, a cool control, outdoor sun exposure, and a desert environment. Each scenario is documented by WBGT, air temperature, relative humidity, and clothing and solar conditions. For every participant-condition trial, the dataset records the following groups of variables:

- **Core physiology (label source only):** rectal temperature (T_{re}) and its change from the trial baseline (Δcore).
- **Cardio-respiratory:** heart rate (HR) and breathing rate (BR).
- **Skin thermal:** temperature at 11 body sites and an insulated-skin estimate.
- **Environmental:** ambient and in-clothing microclimate temperature, humidity, and wind; solar flag and magnitude; work or rest phase; treadmill speed and grade.
- **Individual:** age, sex, body mass, height, body-fat percentage, VO_2peak, acclimation and training status, and clothing ensemble.
- **Sudomotor:** whole-trial sweat loss, a single post-trial scalar (excluded from inputs; see Section IV).

After dropping rows without a core-temperature value, 17,196 of the 17,264 minute-level rows remained, spread over 155 participant-condition trials with a mean duration of about 110 minutes. The dataset citation, DOI, licence terms, and ethics approval are to be inserted by the authors in [20], and the authors should confirm whether earlier studies have used the same data.

B. Label Definition

Risk classes are defined from rectal temperature using conventional occupational action limits, with 38.0 °C as the recommended limit for prolonged heavy work in ISO 7933 [21]:

$$y = \begin{cases} \text{Low} & \text{if } T_{re} < 38.0\,^{\circ}\text{C} \\ \text{Moderate} & \text{if } 38.0 \le T_{re} < 38.5 \\ \text{High} & \text{if } T_{re} \ge 38.5 \end{cases} \tag{1}$$

The 38.0 °C limit is taken from the standard [21] and not from secondary sources. The 38.5 °C boundary is a working choice for a more serious category. Section XIII lists a threshold-sensitivity analysis (for example 38.0/39.0, and acclimatisation-dependent limits) as outstanding work. Fig. 1 summarises the pipeline from sensors to risk class.

C. Class Distribution

After windowing and horizon alignment, 14,103 labelled windows remain: 6,894 Low (48.9%), 4,871 Moderate (34.5%), and 2,338 High (16.6%) (Fig. 2). A uniform random guess over three classes has an expected accuracy of 0.333 and, given these class frequencies, an expected macro-F1 of about

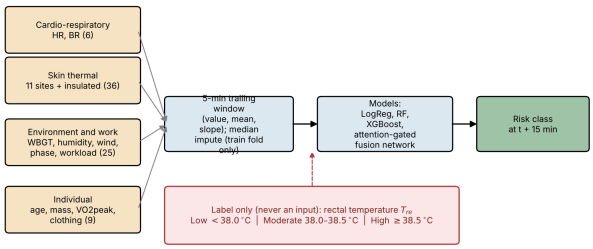


Fig. 1. Pipeline. Four sensor modalities feed a 5-minute trailing window with training-fold imputation; the models predict the risk class 15 minutes ahead. Rectal temperature defines the label and is never an input.

0.319. A classifier that always answers Low has accuracy 0.489 and, because it earns an F1 of 0.657 on Low and zero on the other two classes, a macro-F1 of 0.219. The second reference is the more relevant lower bound because the classes are imbalanced, and we show both throughout. We do not compare against the class balance of other datasets, since we did not measure them.

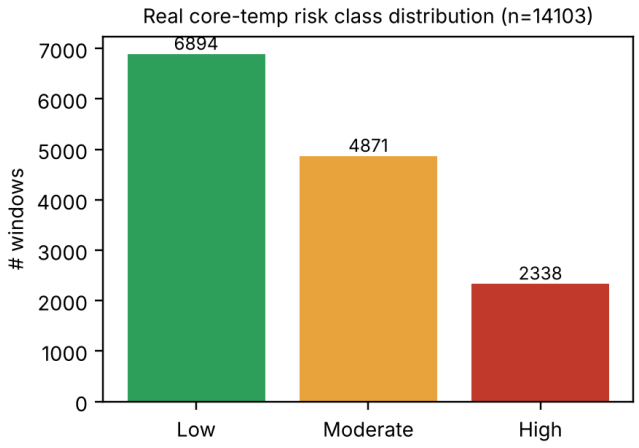


Fig. 2. Class distribution of the core-temperature-derived label over 14,103 windows.

IV. Problem Setup

A. Forecast Horizon and Windows

At each time step the model sees a trailing 5-minute window of sensor features (current value, rolling mean, and rolling slope for continuous channels) and predicts the risk class 15 minutes later. The lead time makes the task a forecast with room for action instead of a same-instant classification. Time is represented as elapsed exposure within the trial, which is a legitimate input because it is known to the system at prediction time.

B. Leakage Rules

Three rules govern which information reaches a model. (i) Core temperature and Δcore are removed because they define the label. (ii) Total sweat loss is removed because it is computed from pre- and post-trial body mass and therefore carries information about the end of the trial that would not be available at prediction time. (iii) Every statistic used for imputation or scaling is computed on the training fold only. These rules follow the general guidance on leakage in predictive modelling [37].

C. Modality Groups

The remaining channels are organised into four modalities that correspond to physically distinct and independently wearable sensor types (Table II). The total is 76 features. The large difference in size between the cardio-respiratory group (6 features) and the others matters for the ablation in Section VIII.

TABLE II
Modality Groups and Feature Counts

Modality	Contents	Features
Cardio-resp.	HR, BR, rolling mean and slope	6
Skin thermal	11 sites + insulated estimate, rolling	36
Environment	Temperature, humidity, wind, WBGT, solar, phase, workload, elapsed time	25
Individual	Age, sex, mass, height, body fat, VO2peak, acclimation, training, clothing	9
Total		**76**

D. Window Features

For each continuous channel the window contributes three quantities: the current value, the mean over the trailing five minutes, and the slope of a least-squares line fitted to the five samples. The mean smooths minute-to-minute sensor noise, and the slope carries direction and speed of change, which matters because a heart rate that is rising steeply has different implications than one that is stable at the same value. Categorical or binary channels such as sex, work or rest phase, and solar flag enter as they are, and elapsed exposure time is a plain counter. As noted in Section II, raw high-frequency motion channels are not used, since they can carry identity information and weaken transfer to new people [14].

E. Missing Data

Missing values occur mostly in breathing rate and heart rate, which are absent for some trials because of sensor availability. They are replaced by the median of the training fold. Median imputation can interact with the cardiac-only ablation, because a model that sees only two variables, partly imputed, may be disadvantaged for reasons unrelated to the sensor type. Section VIII-D reports how much of each channel is missing and repeats the cardiac-only comparison on the subset with complete HR and BR; missing-value indicator features remain among the outstanding analyses in Section XIII.

V. Models

A. Baselines

We compare three standard learners on the flat 76-feature vector: Logistic Regression, Random Forest, and gradient-boosted trees (XGBoost). Each uses class-weighted training to counter the imbalance. These families are chosen because they are common in the heat-strain literature [1]– [3] and because they are strong on tabular data of this kind [29]– [32].

B. Attention-Gated Fusion Network

Each modality $m \in \{\text{cardio}, \text{skin}, \text{env}, \text{ind}\}$ passes through its own small encoder f_m (Linear–ReLU–Dropout–Linear) into a shared embedding space of dimension d:

$$e_m = f_m(x_m) \in \mathbb{R}^d \tag{2}$$

A gate computes per-sample, per-modality weights from the concatenated embeddings and fuses them as a weighted sum, in place of fixed concatenation:

$$\alpha = \text{softmax}(W_g[e_{cardio}; e_{skin}; e_{env}; e_{ind}]) \tag{3}$$

$$z = \sum_m \alpha_m e_m \tag{4}$$

As Fig. 3 shows, the fused vector z feeds a classification head (Linear–ReLU–Dropout–Linear–Softmax) trained with a class-weighted focal loss [25]. Monte Carlo dropout with 25 stochastic forward passes at inference gives a predictive-uncertainty estimate [27].

C. Domain-Adversarial Branch

An optional gradient-reversal layer (GRL) [28] sends z to an auxiliary classifier that predicts participant identity during training. The reversed gradient pushes z toward features from which identity cannot be recovered, a learned counterpart to the context-independent formulation discussed for aPSI in Section II-A [5]. The adversarial weight is increased linearly from zero. Without this ramp, about one random initialisation in three converged to a degenerate predictor that outputs a single class. A random-restart guard, which retrains with a new seed when training-set predictions collapse to one class, removed the failure and reduced the across-fold standard deviation of macro-F1 from 0.107 to 0.038. The number of restarts per fold should be reported with the final results.

D. Implementation

The network is implemented on a custom NumPy automatic-differentiation engine whose gradients were checked against numerical derivatives to an absolute error below 10^{-9} for all custom operators. Training uses Adam with gradient clipping. A reimplementation in a mainstream deep-learning framework, with the NumPy version retained as a cross-check, is planned so that results can be reproduced with standard tooling. Table III lists the configuration that is fixed across experiments.

TABLE III
Experimental Configuration

Item	Setting
Forecast horizon	15 min
Input window	5 min trailing (value, mean, slope)
Windows / trials	14,103 / 155
Imputation	Median, training fold only
Loss	Class-weighted focal (neural); class weights (others)
Optimizer	Adam with gradient clipping; linear GRL ramp-up
MC dropout	25 passes at inference
Calibration	Temperature scaling on participant-level split of training data
Main CV	8-fold GroupKFold by participant

VI. Evaluation Protocol

A. Splitting Protocols

Three protocols are used, each of which keeps held-out data fully separate from training.

1) **8-fold participant-grouped cross-validation.** Participants are divided into 8 groups with GroupKFold so that no person appears in both training and test data of any fold [34]. This protocol supports the main model comparison and all ablations.
2) **Leave-one-participant-out (LOPO).** Forty folds, one participant held out per fold, evaluated with XGBoost for computational reasons. This gives a per-person accuracy for error analysis.
3) **Leave-one-condition-out (LOCO).** Nine folds, one heat-stress scenario held out per fold, model trained on the other eight, evaluated with XGBoost and the full fusion network. Because the same people appear in training (in other scenarios) and in testing, LOCO measures independence from condition and not from person.

B. Metrics, Calibration, and Statistics

Metrics are macro-F1, accuracy, and per-class recall, the last of which is of particular interest for the High class. Calibration is assessed with Expected Calibration Error (ECE) [26] before and after single-parameter temperature scaling, which is fitted on a participant-level calibration split taken from the training data of each fold and never from the held-out fold. Paired comparisons across the 8 folds use the paired t-test and the Wilcoxon signed-rank test. With 8 folds the smallest two-sided p-value attainable by the exact Wilcoxon test is $2/256 \approx 0.0078$, so values of 0.008 reflect the largest possible rank separation, not an unusually small p-value. Because six comparisons are made against the full fusion network, we apply the Holm step-down correction to the Wilcoxon p-values. Fold-wise differences are not independent [33], and for that reason we regard the tests as screening evidence and give the effect sizes alongside them.

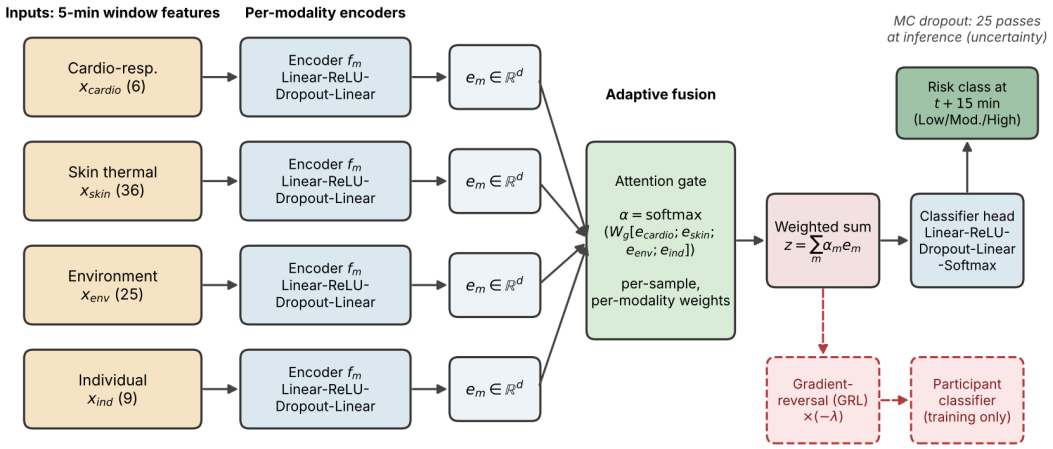


Fig. 3. Architecture of the attention-gated fusion network. Each of the four sensor modalities is encoded separately into a shared embedding space; a softmax gate weights the embeddings per sample and the weighted sum feeds the classifier head. The dashed branch (gradient reversal and participant classifier) is used only in training and is removed in the no-GRL variant.

Macro-F1 averages the per-class F1 scores with equal weight, so a model cannot score well by predicting only the common class. With $K = 3$ classes,

$$F1_{macro} = \frac{1}{K} \sum_{k} \frac{2P_k R_k}{P_k + R_k} \tag{5}$$

where P_k and R_k are the precision and recall of class k. ECE partitions predictions into M confidence bins B_j and averages the gap between accuracy and mean confidence [26]:

$$ECE = \sum_{j} \frac{|B_j|}{n} |acc(B_j) - conf(B_j)| \tag{6}$$

Temperature scaling divides the logits by a single positive scalar T fitted by minimizing negative log-likelihood on the calibration split, which leaves the predicted class unchanged and alters only the confidence [26]. The focal loss with class weight w_y is

$$L_f = -w_y (1 - p_y)^{\gamma} \log p_y \tag{7}$$

where p_y is the predicted probability of the true class and γ controls how strongly easy examples are down-weighted [25].

C. Evaluation Procedure

Each fold, whatever the splitting rule, follows the same sequence so that no step sees held-out data.

1) Assign participants (or scenarios) to training and held-out groups; the held-out group is not touched again until step 5.
2) Compute imputation medians on the training portion and apply them to both portions.
3) Set aside a participant-level calibration subset from the training portion, then train the model on the remainder (neural models use the restart guard).
4) Fit the temperature on the calibration subset, using the same kind of output that will be scored on the held-out group.
5) Predict the held-out group and compute macro-F1, accuracy, per-class recall, and ECE before and after scaling.
6) After all folds, aggregate by mean and standard deviation, then apply the paired tests with Holm correction.

D. Reproducibility

All preprocessing statistics are computed within training folds, and participant identifiers define every split. The random seed, the number of restarts triggered by the collapse guard, and every hyperparameter setting should be listed with the released code so that each number in this paper can be regenerated. The final models were run with three random seeds (0, 1, 2); preprocessing and splits are fixed, so differences between seeds reflect only the randomness of the learners. Where a number in this paper was read from a figure instead of a table, it is given as approximate.

VII. Results A: Model Comparison

Table IV and Fig. 4 give the 8-fold participant-grouped results. XGBoost achieves the highest macro-F1 (0.719 $\pm$ 0.034), followed by the fusion network without the adversarial branch (0.695 $\pm$ 0.046), Random Forest (0.692 $\pm$ 0.040), the full fusion network (0.674 $\pm$ 0.037), and Logistic Regression (0.673 $\pm$ 0.039). Every model with access to all four modalities exceeds both reference levels by a wide margin: the always-Low predictor reaches only 0.219 macro-F1 and 0.489 accuracy, and even the weakest full-input model has accuracy 0.682 (full fusion without attention) or higher.

The always-Low row is computed on the pooled windows and has no fold-wise spread. Recall columns are fold means.

Table V gives the paired tests of the full fusion network against each alternative, with Holm-adjusted Wilcoxon p-values added. The fusion network is worse than XGBoost

TABLE IV
8-FOLD PARTICIPANT-GROUPED RESULTS (MEAN ± STD)

Model	Macro-F1	Acc.	Rec. Mod.	Rec. High
Always-Low (pooled)	0.219	0.489	0.00	0.00
LogReg	0.673±.039	0.697	0.604	0.723
Random Forest	0.692±.040	0.708	0.695	0.688
XGBoost	**0.719±.034**	**0.738**	0.688	0.686
Fusion: cardiac only	0.426±.086	0.469	0.361	0.611
Fusion: no attention	0.660±.056	0.682	0.609	0.697
Fusion: no GRL	0.695±.046	0.715	0.690	0.695
Fusion: full	0.674±.037	0.693	0.640	0.716

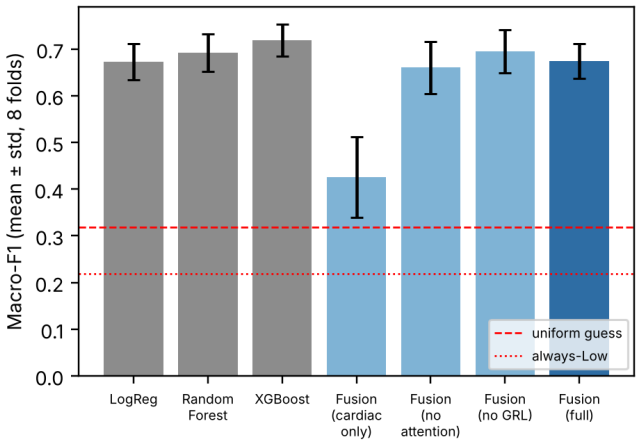


Fig. 4. Macro-F1 for the baselines and fusion variants under 8-fold participant-grouped validation. Dashed line: expected macro-F1 of a uniform guess (0.319); dotted line: always-Low predictor.

by 0.046 macro-F1 (Wilcoxon $p = 0.008$, Holm-adjusted 0.048). It is not distinguishable from Logistic Regression ($p = 0.945$), Random Forest ($p = 0.383$), the no-attention variant ($p = 0.641$), or the no-GRL variant ($p = 0.109$). The comparison with the cardiac-only variant, which differs by $+0.248$, remains significant after correction (Holm 0.048) and is discussed in Section VIII.

TABLE V
PAIRED TESTS, FULL FUSION VS. OTHERS (MACRO-F1, 8 FOLDS)

Comparison	Mean diff.	p (t)	p (Wilc.)	Holm
vs. LogReg	+0.0004	0.958	0.945	1.00
vs. Random Forest	−0.018	0.200	0.383	1.00
vs. XGBoost	−0.046	0.002	0.008	0.048
vs. cardiac-only	+0.248	< 0.001	0.008	0.048
vs. no-attention	+0.014	0.421	0.641	1.00
vs. no-GRL	−0.021	0.063	0.109	0.44

Reference levels give context for these numbers. Table VI places the models against four simple predictors computed from the class frequencies. Every full-input model beats the always-Low predictor by about 0.20–0.25 in accuracy and by 0.45–0.50 in macro-F1. The cardiac-only network is the exception on accuracy: at 0.469 it is below the 0.489 of the always-Low predictor, although its macro-F1 (0.426) is well above 0.219 because it recognises some Moderate and High windows. This shows why accuracy alone is a weak yardstick under imbalance.

TABLE VI
REFERENCE PREDICTORS AND MARGINS OVER ALWAYS-LOW

Predictor	Acc.	Macro-F1	ΔAcc.	ΔF1
Uniform guess (expected)	0.333	0.319	–	–
Always-Low	0.489	0.219	0	0
Always-Moderate	0.345	0.171	–	–
Always-High	0.166	0.095	–	–
XGBoost	0.738	0.719	+0.249	+0.500
Random Forest	0.708	0.692	+0.219	+0.473
LogReg	0.697	0.673	+0.208	+0.454
Fusion: full	0.693	0.674	+0.204	+0.455
Fusion: cardiac only	0.469	0.426	−0.020	+0.207

The fold-to-fold spread is substantial relative to the differences between the better models. Table VII converts the reported standard deviations into approximate 95% intervals for the mean macro-F1 using the t distribution with 7 degrees of freedom. The intervals for XGBoost, Random Forest, Logistic Regression, and the fusion variants without attention or without the GRL overlap heavily, and only the cardiac-only interval is clearly separate. Because fold scores are probably too narrow, these intervals are a reading aid and not a replacement for the corrected tests planned in Section XIII.

TABLE VII
APPROXIMATE 95% INTERVALS FOR MEAN MACRO-F1 (8 FOLDS)

Model	Mean	Interval
LogReg	0.673	0.640–0.706
Random Forest	0.692	0.659–0.725
XGBoost	0.719	0.691–0.747
Fusion: cardiac only	0.426	0.354–0.498
Fusion: no attention	0.660	0.613–0.707
Fusion: no GRL	0.695	0.657–0.733
Fusion: full	0.674	0.643–0.705

Per-class recall shows where the models struggle. Recall on the Low class is highest (0.73–0.80 for the full-input models), recall on the High class is about 0.69–0.72, and recall on the Moderate class is lowest (0.60–0.70). This ordering fits the structure of the label: Moderate is a band between two thresholds and is therefore the class most exposed to small errors in the forecast temperature, while Low and High lie at the extremes. For an alerting application the practical consequence is that confusion between neighbouring classes is more likely than confusion between Low and High; the confusion matrices in Section XI-A quantify this.

Apart from the cardiac-only variant, High-class recall (Fig. 5) lies in a narrow band from 0.69 to 0.72, with Logistic Regression (0.723) and the full fusion network (0.716) at the

top and XGBoost (0.686) and Random Forest (0.688) just below. The cardiac-only variant is lower at 0.611. The data therefore do not show that fusion consistently improves recall on the High class; the differences among the better models are within what fold-to-fold variation could produce.

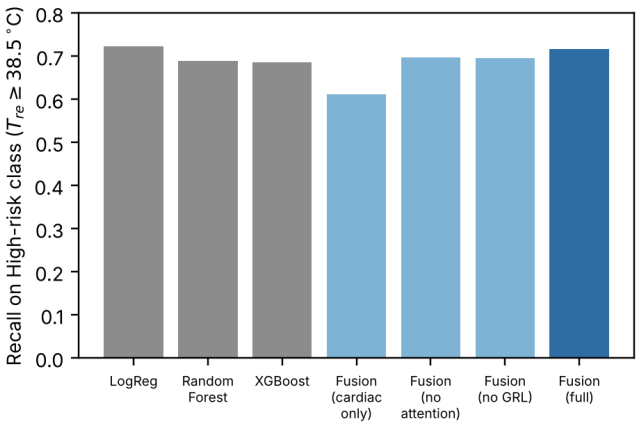


Fig. 5. Recall on the High-risk class ($T_{re} \geq 38.5\,^{\circ}$C) by model.

VIII. Results B: Modality and Architecture Studies

A. Cardiac-Only Input

Restricting the fusion network to heart rate and breathing rate lowers macro-F1 from 0.674 to 0.426, an absolute drop of 0.248, with High-class recall falling from 0.716 to 0.611. The paired difference is among the largest effects in the study and survives correction for multiple testing (Table V). This indicates that skin-thermal, environmental, or individual information adds predictive value that heart and breathing rate alone do not supply when the network is trained as described.

Three features of the design limit how far this finding can be pushed. The cardiac group has 6 inputs against 76 for the full set, so part of the gap may reflect input dimensionality alone. Heart rate and breathing rate are also the channels with missing values, so imputation could have pulled performance down. Finally, in the fusion experiments only one single-modality configuration was run. We therefore ran the missing controls with XGBoost (Section VIII-D). They reproduce the weakness of the cardiac group (macro-F1 0.417 with heart and breathing rate alone) and, on the subset of windows where both channels were actually recorded, narrow the cardiac-only deficit from about 0.30 to about 0.15 without removing it. Part of the drop in the fusion experiment is therefore a missing-data effect, and a substantial part is not. Feature-count-matched controls are still outstanding, so we describe the result as a large and consistent difference whose cause is only partly isolated.

B. Attention Gate

Replacing the learned attention gate by fixed equal weights lowers macro-F1 by 0.014 relative to the full model ($p = 0.641$ by Wilcoxon, 0.421 by t-test) and by 0.035 relative to the no-GRL variant. These differences are small and neither is statistically distinguishable from zero in this experiment, so we do not claim that attention improves accuracy. The attention weights themselves are examined in Section XI-C. They differ between scenarios and risk classes but are not stable across random seeds, which is one more reason not to credit the gate with the accuracy results; the aPSI comparison discussed in Section II-A remains the natural point of reference for what a context-dependent weighting should achieve.

C. Domain-Adversarial Branch

Removing the GRL gave the best fusion variant (macro-F1 0.695, against 0.674 with the GRL), although the difference is not significant ($p = 0.109$). Together with the training collapse noted in Section V, the evidence suggests that participant-adversarial training adds cost without benefit on this dataset. An alternative worth testing is to use scenario, instead of participant, as the domain label, and to align the adversarial objective with the LOCO setting.

D. Single-Modality and Leave-One-Modality-Out Runs (XGBoost)

The fusion experiments isolated only the cardiac group, so we ran the missing controls with XGBoost, using the same grouped folds and a single seed. Table VIII gives the result. Skin-thermal features alone reach a macro-F1 of 0.684, close to the 0.716 obtained with all 76 features, and removing them costs 0.107, whereas removing any one of the other three groups changes macro-F1 by 0.03 or less. Environment alone reaches 0.571, heart and breathing rate alone 0.417, and the nine static descriptors of the individual 0.285, which is below the uniform-guess level (0.319) and unsurprising, because those values are constant within a person and cannot indicate how a trial is evolving. Read together, these runs say that for this model the skin-thermal channels carry most of the unique information, the environment channels add a smaller share, and heart and breathing rate add little once skin temperature is available.

Two cautions apply. First, the groups differ in size (6, 36, 25, and 9 features), and although the ordering is not strictly monotone in feature count, a count-matched control has not been run. Second, the cardiac group is the one most affected by missing data: heart rate is absent in 30.8% of windows and breathing rate in 44.5%, with breathing rate missing for the whole of 64 of the 155 trials and heart rate missing in part of 60 trials. On the 7,704 windows (22 participants) where both are present, XGBoost with cardiac features alone reaches a macro-F1 of 0.527, against 0.673 for all features on the same subset. Missing data therefore explain part of the weak cardiac-only score, but a gap of about 0.15 remains. The results come from one seed and one learner, and the modalities are correlated, so they are best treated as a guide for the sensor studies listed in Section XIII.

TABLE VIII
XGBOOST MODALITY RUNS (8 FOLDS, SEED 0; MEAN ± SD)

Features used	No.	Macro-F1	Acc.	Rec. High
All (reference)	76	0.716±.038	0.736	0.682
Cardio-resp. only	6	0.417±.056	0.449	0.635
Skin thermal only	36	0.684±.037	0.702	0.682
Environment only	25	0.571±.057	0.607	0.548
Individual only	9	0.285±.050	0.359	0.142
All but cardio-resp.	70	0.714±.041	0.734	0.672
All but skin thermal	40	0.609±.041	0.649	0.524
All but environment	51	0.687±.037	0.709	0.657
All but individual	67	0.707±.033	0.730	0.669
Windows with both HR and BR recorded (7,704; 22 participants)				
All features	76	0.673±.042	0.725	–
Cardio-resp. only	6	0.527±.081	0.592	–

IX. RESULTS C: GENERALIZATION

A. Leave-One-Participant-Out

Fig. 6 shows XGBoost accuracy for each of the 40 held-out participants. Mean accuracy is 0.736 ± 0.072, ranging from 0.599 to 0.876, and every participant is above the 0.489 accuracy of the always-Low reference computed over all windows. Because a given participant's own class mix can differ from the pooled one, a participant-specific majority baseline is the fairer comparison and is listed among the outstanding analyses. The per-participant bars in Fig. 6 are the real fold results of the 40-fold run.

Per-participant accuracy correlates with the share of that participant's windows labelled High at $r = 0.30$ (Fig. 7). With $n = 40$ the 95% confidence interval for this correlation, from the Fisher transformation, is approximately -0.01 to 0.56, a weak positive association is the strongest statement the data support. Age correlates at $r = 0.19$ (interval about -0.13 to 0.47) and aerobic fitness (VO_2peak) at $r = 0.004$ (about -0.31 to 0.32); neither is distinguishable from zero. We note, without over-reading a single modest correlation, that a weak link between High-risk prevalence and model accuracy is broadly consistent with the individual-factor literature discussed in Section II-D, which argues that who a worker is — not only where they work — shapes measured strain [7]; it is also consistent with simpler explanations such as class imbalance within a person's own trial mix, and the data here cannot separate the two.

B. Leave-One-Condition-Out

Fig. 8 gives accuracy for each scenario when it is withheld from training. The mean is 0.682 ± 0.078 for XGBoost and 0.650 ± 0.094 for the full fusion network. The scenario labels C1–C3 and C6–C11 follow the dataset's own condition identifiers, so nine scenarios are shown although the labels run to 11. For every held-out scenario both models are above the 0.489 pooled majority level except condition 8 (indoor hot factory, full protective clothing), where accuracy is 0.43–0.51 and therefore close to or below that level. Heavy protective

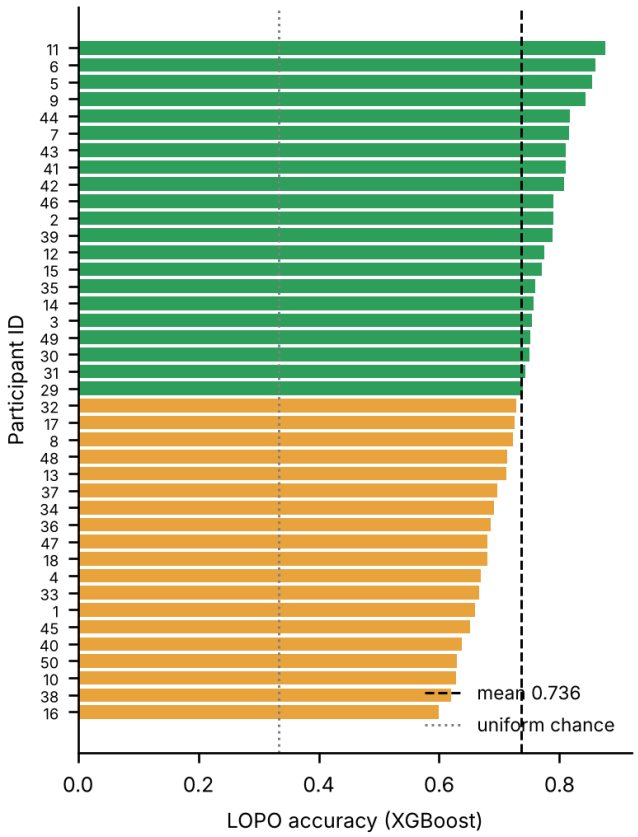


Fig. 6. LOPO accuracy (XGBoost) for all 40 participants. Dashed line: mean (0.74); dotted line: uniform chance.

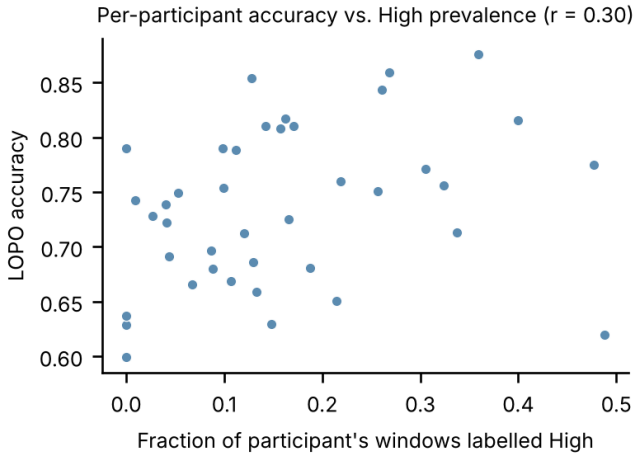


Fig. 7. Per-participant LOPO accuracy against the fraction of that participant's windows labelled High ($r = 0.30$).

ensembles evidently change the relation between the measured signals and core-temperature rise enough to challenge a model trained mostly on lighter-clothing scenarios, which is a concrete caution for deployment and lines up closely with the protective-equipment finding in [8], where ambient heat and PPE use together moved ear temperature, heart rate, and skin temperature beyond what either factor produced alone. Scenario-specific majority baselines, which would sharpen this conclusion, still have to be added.

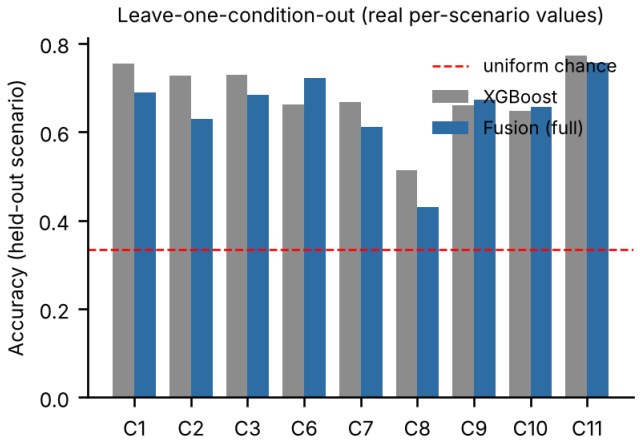


Fig. 8. LOCO accuracy: trained on eight scenarios and tested on the ninth. Dashed line: uniform chance. Labels follow the dataset's condition identifiers.

Table VIII collects the generalization results with intervals for the mean across held-out units (participants for LOPO, scenarios for LOCO) using the t distribution. The LOPO interval is narrow (0.713–0.759) because it rests on 40 participants, while the LOCO intervals are wide (0.622–0.742 for XGBoost and 0.578–0.722 for the fusion network) because only nine scenarios are available. The 0.032 gap between the two models under LOCO lies well inside these intervals, so the data do not separate them.

TABLE IX
GENERALIZATION SUMMARY (ACCURACY)

Protocol / model	Units	Mean ± SD	95% interval
LOPO, XGBoost	40	0.736 ± 0.072	0.713–0.759
LOCO, XGBoost	9	0.682 ± 0.078	0.622–0.742
LOCO, full fusion	9	0.650 ± 0.094	0.578–0.722

The LOCO result is condition-independent but not person-independent. A combined hold-out, in which a scenario and the participants who took part in it are withheld together, would be a stricter test where the design of the dataset allows it.

X. RESULTS D: CALIBRATION

Table IX and Fig. 9 give ECE before and after temperature scaling. Scaling improved calibration for every tree and linear baseline (XGBoost 0.060 to 0.052, Logistic Regression 0.083 to 0.052, Random Forest 0.081 to 0.059). For three of the four fusion variants, however, it made calibration worse (full fusion 0.085 to 0.170, no attention 0.088 to 0.174, no GRL 0.080 to 0.165), and for the cardiac-only variant it gave a small improvement (0.110 to 0.099).

TABLE X
EXPECTED CALIBRATION ERROR, RAW VS. TEMPERATURE-SCALED

Model	ECE (raw)	ECE (scaled)
LogReg	0.083	0.052
Random Forest	0.081	0.059
XGBoost	0.060	0.052
Fusion: cardiac only	0.110	0.099
Fusion: no attention	0.088	0.174
Fusion: no GRL	0.080	0.165
Fusion: full	0.085	0.170

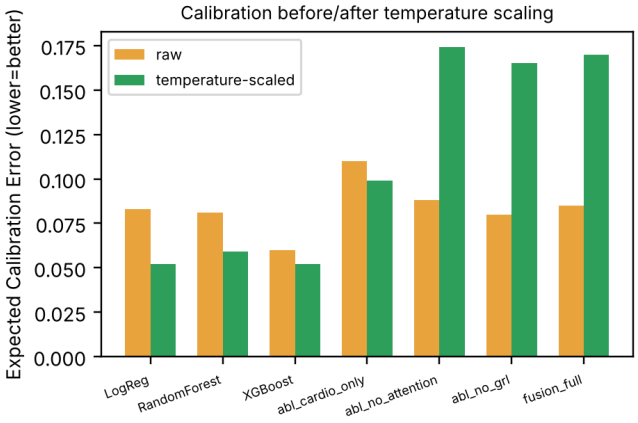


Fig. 9. ECE before and after temperature scaling by model.

A doubling of ECE after fitting a single parameter on held-out data is unusual enough that an implementation or protocol cause should be excluded before a modelling explanation is accepted. Candidates are: (i) a mismatch between the outputs used to fit the temperature (single forward pass) and those used at test time (MC-dropout average); (ii) the effect of class-weighted focal loss on the output probabilities, which does not target calibration and can leave scores that a single temperature cannot correct; and (iii) the size of the participant-level calibration split. Vector scaling and Dirichlet calibration [26], together with Brier score, negative log-likelihood, and reliability diagrams, are the planned next steps. Until these checks are complete, the calibration result for the fusion network should be treated as unresolved and not as a property of the architecture.

XI. RESULTS E: ERROR ANALYSIS, FORECAST TRAJECTORIES, AND SANITY CHECKS

The analyses in this section were added after the main experiments and use out-of-fold (OOF) predictions: every

window is scored by a model that never saw its participant, and the predictions from the eight folds are pooled. All values come from a re-run of the same configuration with three random seeds (0, 1, 2) for Logistic Regression, Random Forest, XGBoost, and the full fusion network; Section XI-D explains how this re-run relates to Table IV.

A. Extended Metrics and Confusion Matrices

Table XI adds chance-corrected and probability-based measures to the accuracy and macro-F1 used so far. Cohen's kappa and the Matthews correlation coefficient [35] discount the agreement expected by chance, which matters because the Low class holds nearly half of the windows; the Brier score and the area under the precision–recall curve for the High class judge the probabilities and not only the label. XGBoost is best on every row, with a Matthews coefficient of 0.576 against 0.533 for the fusion network, and Random Forest and Logistic Regression fall in between or level with the fusion network. Pooled macro-F1 differs slightly from the fold means in Table IV because pooling weights every window equally, whereas a fold mean weights every fold equally.

TABLE XI
EXTENDED METRICS ON POOLED OUT-OF-FOLD PREDICTIONS (MEAN ± SD OVER THREE SEEDS)

Model	Macro-F1	Bal. acc.	κ	MCC	AUPRC High	Brier
XGBoost	0.722±.004	0.720	0.575	0.576	0.792	0.358
Random Forest	0.698±.001	0.705	0.540	0.545	0.738	0.391
LogReg	0.682±.000	0.698	0.526	0.528	0.689	0.404
Fusion (full)	0.686±.011	0.699	0.531	0.533	0.693	0.401

Per-class precision, recall, and F1 are in Table XII. Both models handle the Low class best. The Moderate class is the weakest for both, as expected for a band between two thresholds.

TABLE XII
PER-CLASS PRECISION (P), RECALL (R), AND F1 (SEED MEANS)

	XGBoost			Fusion (full)		
Class	P	R	F1	P	R	F1
Low	0.848	0.793	0.820	0.858	0.759	0.806
Moderate	0.615	0.685	0.649	0.586	0.624	0.604
High	0.715	0.682	0.698	0.593	0.715	0.649

Fig. 10 shows that nearly all errors fall between neighbouring classes. For XGBoost, 0.9% of Low windows are forecast as High and 2.1% of High windows as Low; for the fusion network the corresponding figures are 2.0% and 1.4%. The Moderate class is the least stable, with 20.0% forecast as Low and 12.0% as High by XGBoost, against 16.4% and 21.2% for the fusion network. The two models therefore make different trade-offs on the High class. XGBoost is more cautious; the fusion network moves more Moderate windows into High, which raises its recall to 0.715 but lowers its precision to 0.593. The ranking quality for High-versus-rest agrees with this picture (AUROC 0.94 for XGBoost and 0.91 for the fusion network, Fig. 11).

B. Forecast Trajectories

Fig. 12 shows three held-out trials. To avoid picking flattering examples we used a fixed rule: among the 91 trials that contain all three classes, trials were ranked by the fusion network's window accuracy, and those at the 25th, 50th, and 75th percentiles are shown (accuracy 0.64, 0.73, and 0.79). In each panel the black line is the measured rectal temperature, which the models never see; the coloured curves are the predicted probability that the state 15 minutes later is High; and the strips compare the true class with each model's forecast, one cell per forecast issued.

The curves behave as a forecaster should. In the upper-quartile trial (participant 15, scenario C1) the predicted probability of High climbs steeply at about minute 38, while measured T_{re} first reaches 38.5 °C at minute 55, so the forecast leads the event by roughly the intended 15 minutes. The errors are also easy to see. In the lowest-quartile trial (participant 38, scenario C11) both models lag when T_{re} is rising through the Moderate band, and the fusion network forecasts Low for roughly 15 minutes while the true class is Moderate and T_{re} is still about 38.2–38.4 °C. Most mistakes are of this boundary-timing kind.

The same question can be asked over all trials (Table XIII, seed 0, no smoothing). Of the 155 trials, 63 reached 38.5 °C. XGBoost issued a High forecast before the crossing in 58 of these 63 (92%), with a median lead of 15.0 minutes (interquartile range 9.8–18.3); the fusion network did so in 55 (87%), with a median lead of 14.0 minutes (8.0–19.0). Requiring three consecutive High forecasts lowers these counts to 55 and 53 and the median leads to 13.5 and 13.0 minutes. The price is a high false-alarm rate at the default decision rule: among the 92 trials that never reached 38.5 °C, 38 (XGBoost) and 50 (fusion) received at least one High forecast, and 32 and 42 received a persistent one. A deployed alert would need a tuned threshold, smoothing, and calibrated probabilities, none of which is established here.

TABLE XIII
EVENT-LEVEL VIEW OF THE FORECASTS (155 HELD-OUT TRIALS, SEED 0)

	XGBoost	Fusion (full)
Trials reaching 38.5 °C (of 155)	63	63
High forecast before crossing	58 (92%)	55 (87%)
Median lead, min (IQR)	15.0 (9.8–18.3)	14.0 (8.0–19.0)
Three consecutive High forecasts before crossing	55	53
Median lead, persistent, min	13.5	13.0
Trials never reaching 38.5 °C	92	92
with any High forecast	38	50
with a persistent High forecast	32	42

C. Fusion Weights

The fusion network's gate assigns each window a weight to each modality (Fig. 13). Averaged over the held-out windows of seed 0, the weights are 0.25 for heart and breathing rate, 0.30 for skin temperature, 0.27 for the environment, and 0.17

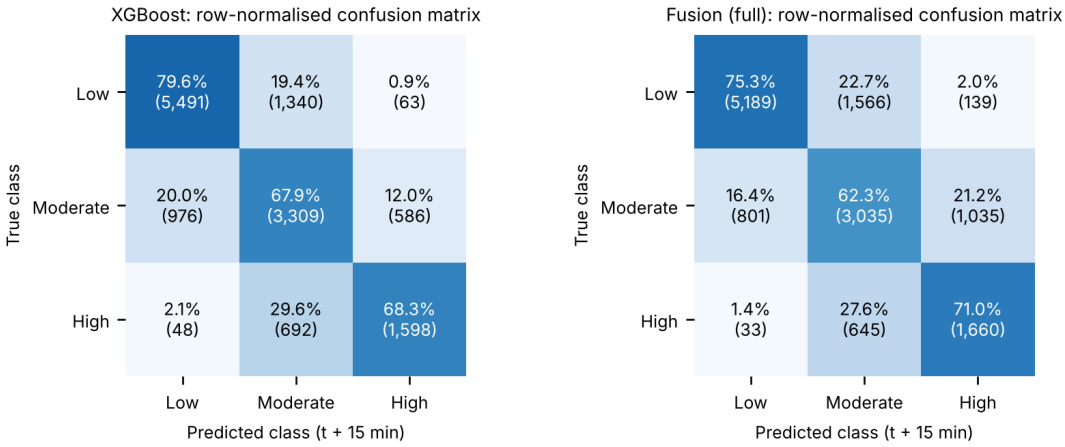


Fig. 10. Row-normalised confusion matrices of the pooled out-of-fold predictions (seed 0); counts in brackets.

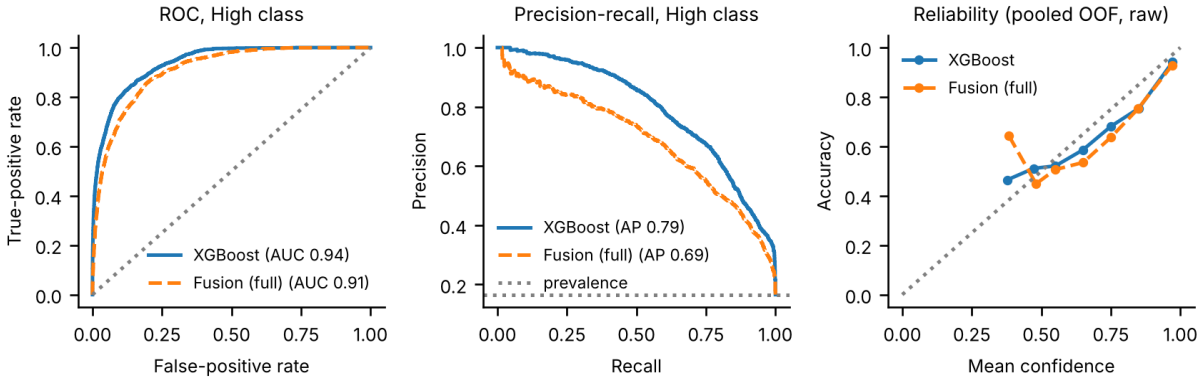


Fig. 11. High-versus-rest ROC and precision–recall curves, and a reliability diagram of the raw predicted confidence, for XGBoost and the full fusion network (pooled out-of-fold predictions, seed 0).

for the individual descriptors. They vary with context. By scenario, the skin-thermal weight ranges from 0.19 (C8) to 0.56 (C2) and the individual weight from 0.07 (C6) to 0.30 (C8). By true risk class, the weight on heart and breathing rate is highest for Low windows (0.33) and lowest for High windows (0.16), while the weight on the individual descriptors rises from 0.12 (Low) to 0.25 (High). In C8, the scenario with the lowest accuracy, the gate gives its largest weight to the individual modality, but with one seed and no causal test we do not read anything into that.

The weights should not be over-interpreted, for two reasons. First, they are not stable across random seeds (Table XIV): the environment weight averages 0.27, 0.17, and 0.30, and the skin-thermal weight 0.30, 0.31, and 0.22, so the sensor that the gate favours depends on the initialisation. Only the individual modality is consistently the lowest (0.17–0.20). Second, the gate weights learned embeddings and not raw inputs, and the embeddings have no common scale, so a weight is not a measure of sensor importance. This agrees with the earlier finding that replacing the gate by equal weights changes macro-F1 by only 0.014 (Section VIII-B), and the XGBoost modality runs of Section VIII-D give a more direct, if still limited, view of which channels matter.

TABLE XIV
MEAN ATTENTION WEIGHT PER MODALITY, BY RANDOM SEED

Seed	Cardio-resp.	Skin thermal	Environment	Individual
0	0.252	0.304	0.272	0.172
1	0.320	0.307	0.169	0.204
2	0.291	0.218	0.300	0.191

D. Training Behaviour and Sanity Checks

Because the numbers above should not be the product of a pipeline artefact, we ran four checks.

Label permutation. We shuffled the class labels across windows and ran the same grouped cross-validation with XGBoost, five times [36]. Macro-F1 was 0.326 $\pm$ 0.005 (range 0.322–0.334) and accuracy 0.363 $\pm$ 0.006, which is the uniform-guess level (0.319 and 0.333) and far below the 0.716 obtained with the true labels. This shows that the pipeline does

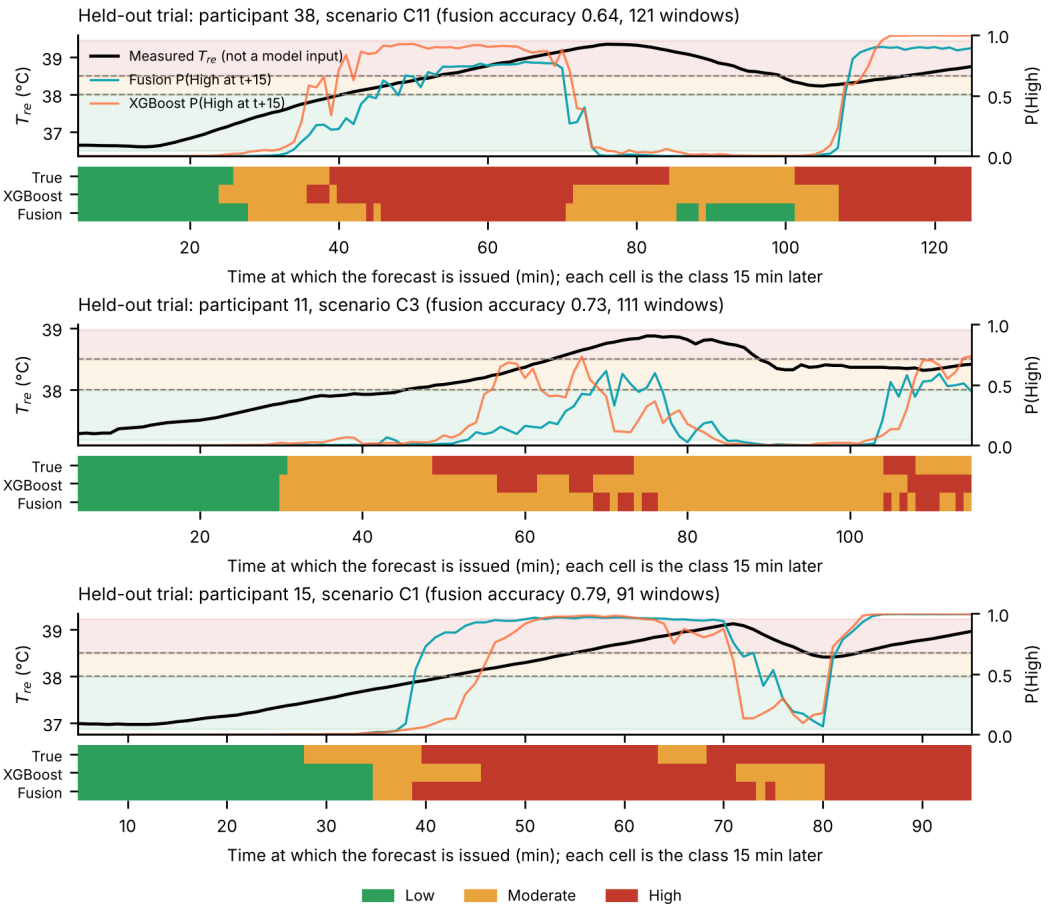


Fig. 12. Forecast trajectories for three held-out trials, selected by the fixed rule in the text (25th, 50th, and 75th percentile of fusion accuracy). Measured T_{re} is shown for reference only. The probability plotted at time t refers to the state at $t + 15$ min, so it should rise about 15 minutes before T_{re} crosses a threshold. Strips: true class, XGBoost forecast, and fusion forecast, each 15 minutes ahead of the time on the axis.

not produce high scores from structure unrelated to the labels. It is a pipeline-level control and not a proof that no leakage exists; the rules in Section IV-B remain the main safeguard.

Seed stability and reproduction of Table IV. Over three seeds the fold-mean macro-F1 of XGBoost was 0.717 ± 0.004 (Table IV: 0.719), that of Random Forest 0.692 ± 0.001 (0.692), that of Logistic Regression 0.674 (0.673, deterministic), and that of the full fusion network 0.677, 0.673, and 0.693 (mean 0.681 against 0.674 in Table IV). XGBoost is highest in every seed; the order of the other three varies. The neural network is not bit-for-bit reproducible across software versions, so fold-level values differ from the original run, but the aggregates agree to within 0.007.

Training curves. Fig. 14 shows that the running training accuracy rose from 0.63 in the first epoch to 0.86 at epoch 100 (dropout active), and that the loss fell from 1.14 to 1.05. The loss includes the adversarial participant term, which is deliberately kept high, so its decline is modest. Held-out accuracy is about 0.70, so the network fits its training windows noticeably better than unseen participants; a gap of roughly 0.15 is plausible with 76 features and about 35 training participants, and regularisation and early stopping were not tuned.

Collapse guard. The random-restart guard did not trigger in any of the 24 final runs (three seeds, eight folds).

XII. Discussion

Simple models are strong. Gradient-boosted trees and logistic regression are competitive with, and here modestly stronger than, an attention-gated fusion network on aggregate accuracy. The same pattern is reported for other tabular prediction problems [29]– [32] and for heat-strain forecasting in particular [1], [3]. In our data the relevant difference is that XGBoost leads the full fusion network by 0.046 macro-F1 and that the difference survives correction, while Logistic Regression and the fusion network are indistinguishable.

What fusion has and has not shown. The case for fusion rests on two observations: a large drop when only cardiac data are used, and High-class recall that is similar to the best baseline. The first has confounds, which Section VIII-D only partly resolves. The second is not a win: Logistic Regression reaches a recall that is essentially equal (about 0.72 against 0.716), and XGBoost trails by about 0.03 (0.686

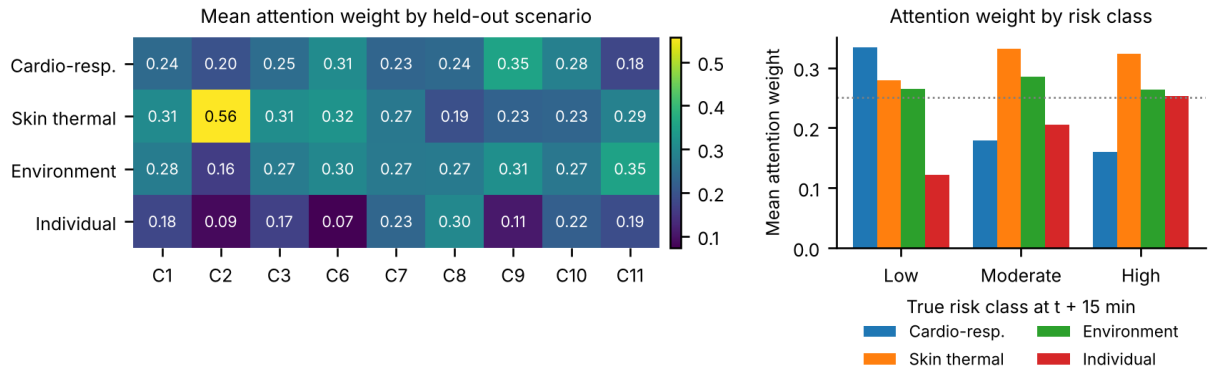


Fig. 13. Mean attention (fusion) weight per modality for the full fusion network (seed 0), grouped by the scenario of the held-out windows (left) and by their true risk class (right). The dotted line marks equal weighting (0.25).

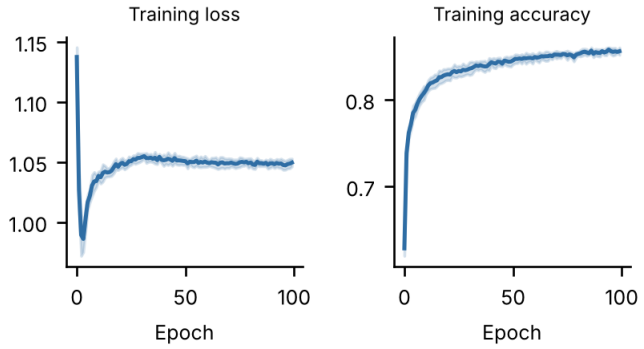


Fig. 14. Training loss and running training accuracy of the full fusion network (mean ± SD over the eight folds, seed 0).

against 0.716). Where the fusion network could still earn its place is for robustness, because an attention gate can, in principle, down-weight a failed modality. That hypothesis has not been tested here and should be, using test-time sensor dropout, noise injection, and training with modality dropout.

Why not compare with regression studies directly. Earlier studies report mean absolute errors below 0.05 °C for core or tympanic temperature [2], [1]. Those targets are continuous and strongly autocorrelated, so a simple persistence forecast over a short horizon already carries a small error, and the studies differ from ours in split design: Seo et al.'s strongest reported result evaluates within the same population used for training, and the error is reported against a measured value whose own minute-to-minute change is small relative to the error itself, whereas our label is a discrete class predicted 15 minutes ahead for people the model has never trained on. Our macro-F1 of 0.72 for a three-class label under person-independent evaluation is a different quantity and the two should not be ranked against each other on a single shared scale. A regression variant that predicts T_{re} at +15 minutes without T_{re} as an input, reported with MAE against a persistence baseline, would allow a like-for-like comparison and is listed as outstanding work.

Operating points. A recall of about 0.72 on the High class means that roughly 28 of every 100 windows that precede a High state go undetected at the default decision rule. Which trade-off is acceptable depends on the cost of missed events relative to false alarms, and that cost differs between a crew supervisor receiving a few alerts per shift and an automated cooling system. Because the models output class probabilities, the threshold for the High class can be moved, and the right place to set it is a precision–recall and false-alarms-per-hour analysis, which requires the calibrated outputs discussed in Section X. The confusion matrices in Section XI-A show that the two models already sit at different operating points: XGBoost is the more cautious (High precision 0.715 at recall 0.682), whereas the fusion network buys a recall of 0.715 at a precision of 0.593. The event-level view in Section XI-B adds the cost side, since many trials that never reached 38.5 °C still received a High forecast at the default decision rule. For that reason the calibration problem is more than a technical detail: an alert threshold set on miscalibrated probabilities will not behave as intended.

Sensor practicality. The four modalities differ in how easily they can be deployed. Heart and breathing rate can come from a chest patch or a strap, environmental inputs from site instruments, and individual descriptors from onboarding records. Skin-temperature measurement at eleven body sites is not realistic outside the laboratory, so the more useful question for hardware design is how few sites, or which cheaper proxies, retain most of the information. The XGBoost modality runs in Section VIII-D narrow the question, because the skin-thermal channels carry most of the unique information, but they cannot say which sites matter; the sensor-set and skin-site studies in Section XIII are intended to.

Deployment meaning. Two findings have practical value even though the architecture does not. First, every one of 40 unseen workers and eight of nine unseen scenarios were

predicted above the pooled majority level, which supports the view that physiological and environmental sensing carries information beyond what a fixed label rule would give. Second, performance fell for the protective-clothing scenario, which suggests that clothing ensemble should be an explicit factor in any deployed alerting system, a point the review of occupational-indicator standards in [4] makes from a different angle when it treats clothing insulation (ISO 9920) as a mandatory input to any physics-based strain model rather than an optional refinement. Consistent with the evidence that self-reported symptoms track physiological strain poorly [9] and that individual factors are under-weighted in standard indices [7], the individual modality is kept in the model, but its separate contribution has not yet been isolated and cannot be assessed from the present ablations.

XIII. Limitations and Future Work

The study uses 40 participants in 9 laboratory scenarios. Field validation on larger and more varied occupational groups, including women workers as highlighted in [7], is needed before any deployment claim. The analyses below are still needed to complete the evaluation; where part of an item has already been done in this version, the item says so.

1) **Base checks.** Done in this version: a missingness summary and a rerun on windows with complete HR and BR (Section VIII-D), and a label-permutation control (Section XI-D). Still open: missing-value indicator features, per-condition and per-participant majority baselines, and a corrected calibration protocol (Section X).
2) **Stronger and fairer comparisons.** LightGBM, an MLP, a late-fusion MLP, a sequence model (LSTM or temporal CNN) on raw 20-minute windows, and a tabular transformer [31]; nested or fixed-inner-split tuning with a hyperparameter table; stratified grouped CV repeated over several partitions of the participants (three random seeds of the learners are already reported in Section XI-D), with bootstrap confidence intervals over participants and corrected repeated-CV tests [33].
3) **Field-index comparison.** WBGT thresholds, PHS-predicted core temperature [21], and PSI [5] use T_{re}, so it must be shown as an oracle, and a lower bound using only time and scenario should be given. A direct comparison with the adaptive aPSI formulation (Eq. 4–5 of [5]), computed on the same PROSPIE trials and converted to the same three-tier risk scale used here, would let a fixed physiologically-adapted index and a learned fusion network be judged on identical held-out data for the first time.
4) **Modality and sensor studies.** Single-modality and leave-one-modality-out runs are reported for XGBoost only (Section VIII-D). Still open: the same runs for the fusion network, feature-count-matched controls, realistic sensor sets (chest patch; chest patch plus two skin sites; wrist heart rate plus ambient), skin-site selection, and SHAP [32] for XGBoost.
5) **Robustness.** Test-time sensor dropout, noise injection (for example ± 3 bpm on HR and $\pm 0.3\,^{\circ}\mathrm{C}$ on skin temperature), 1-minute versus 5-minute sampling, and modality-dropout training.
6) **Forecast variants and alerting.** Horizons of 5, 10, 15, and 30 minutes; a regression target; threshold sensitivity; ordinal loss; and alert-style metrics such as High-class precision, recall, and PR-AUC, false alarms per hour, lead time, and event-level detection, with uncertainty-versus-error and abstention analysis for MC dropout.
7) **Climate-scenario robustness.** Following the approach of [6], who project how safe working duration under PHS changes across representative concentration pathways and future periods, a forward-looking check of whether the present classifier's accuracy and calibration hold under warmer baseline conditions than those sampled by PROSPIE's nine scenarios would clarify whether the model is tied to the specific thermal range it was trained on or generalises to the harsher conditions climate projections anticipate.

Responsible Use

A forecast of this kind should support, and not replace, established measures such as acclimatization programmes, work-rest scheduling, hydration, and supervision under the applicable standards. The models were developed on a small laboratory sample and have not been checked for differences in performance across sex, age band, acclimatization, or fitness; subgroup analyses are needed before any use that affects individual workers, and the labels reflect core-temperature thresholds that individual workers can exceed or tolerate differently.

Threats to Validity and Decision Rules

Several threats apply to the present numbers. The sample is small, so estimates carry wide intervals [33]. The tuning protocol is not yet documented to the standard needed to rule out optimistic selection of settings. Eight folds from one partition of the participants give a single realization of the split, and repeated stratified grouped partitions would show how sensitive the ranking of models is to it. The label thresholds are a convention and not a physiological boundary, so conclusions about class recall depend on them. Finally, the laboratory trials followed controlled protocols, and field work involves irregular activity, intermittent sensor contact, and varied clothing.

To keep the interpretation tied to evidence, we state in advance how the outstanding analyses would change the conclusions (Table X).

Finally, the PhysioNet working-memory physiological signal database [19], used elsewhere for cross-subject pretraining [14], could supply unlabelled data for pretraining the cardio-respiratory and skin encoders, provided the adversarial domain label is the dataset of origin and not participant identity, to avoid the instability seen here. A temporal encoder in place of rolling-window features is a further direction consistent with [1]. Total sweat loss, an important strain indicator, was unusable as a real-time feature, and a dataset with continuous sweat-rate sensing would allow it to be included, which would

TABLE XV
PLANNED DECISION RULES

If the rerun shows	Then the paper will
Fusion wins under sensor dropout	Present fusion as the robust option
Fusion loses everywhere	Keep trees as the recommended model
Corrected calibration works for fusion	Report it as a contribution
Cardiac-only on complete HR/BR (checked with XGBoost: 0.527 against 0.673 for all features)	Claim kept, with a smaller gap of about 0.15
Ranking changes across repeated splits	Report ranks with intervals only

also let the competing core-temperature and sweat-loss failure modes identified by [6] be modelled jointly rather than through core temperature alone.

XIV. CONCLUSION

We presented a leakage-safe benchmark for forecasting occupational heat-strain risk 15 minutes ahead, using real rectal-temperature labels from PROSPIE and evaluating on unseen participants and unseen scenarios. Gradient-boosted trees (macro-F1 0.719) matched or exceeded an attention-gated fusion network (0.674), and both were far above an always-Low reference (0.219). The large drop under cardiac-only input suggests that other sensor groups carry information, but feature count and missing data must be controlled before this is established. LOPO and LOCO accuracy were well above the pooled majority level for nearly all participants and scenarios, with protective-clothing work as the clear exception. Temperature scaling helped the baselines and hurt the fusion network. The adversarial branch gave no benefit. Errors fall mostly between neighbouring classes, label-permuted controls fall to chance, and the attention weights are not stable across random seeds. Set against recent work on feature-importance-driven heat-strain forecasting [1], [2], adaptive physiological indexing [5], a structured review of heat-stress and heat-strain indices [4], and climate-projected labour-capacity loss [6], the present results are best read as showing where a learned, multimodal model is clearly better than a fixed-form index — unseen-worker generalisation above chance — and where it is not yet shown to be better, namely aggregate accuracy against simple tree ensembles and calibrated confidence. The outstanding analyses in Section XIII define what remains to be done for a complete benchmark.

DECLARATIONS

Data and code availability: [repository link and licence to be inserted]. **Funding:** [to be inserted]. **Author contributions:** [to be inserted]. **ORCIDs:** [to be inserted]. **Ethics:** the data are from the public PROSPIE dataset; the source study's ethics approval is to be cited in [20].

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